

Packet Tracer - TCP and UDP Communications

Objectives

- Part 1: Generate Network Traffic in Simulation Mode
- Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding TCP and UDP in detail. Packet Tracer simulation mode provides you the ability to view the state of different PDUs as they travel through the network.

Packet Tracer Simulation mode enables you to view each of the protocols and the associated PDUs. The steps outlined below lead you through the process of requesting network services using various applications that are available on a client PC. You will explore the functionality of the TCP and UDP protocols, multiplexing, and the function of port numbers in determining which local application requested the data or is sending the data. Packet Tracer will not score this activity.

Instructions

Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

- **Step 1:** Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following task to reduce the amount of network traffic viewed in the simulation.

- a. Click MultiServer and click the Desktop tab > Command Prompt.
- b. Enter the ping -n 1 192.168.1.255 command. You are ping the broadcast address for the client LAN. The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.
- c. Close the MultiServer window.

- **Step 2:** Generate web (HTTP) traffic.

- a. Switch to Simulation mode.
- b. Click HTTP Client and open the Web Browser from the desktop.
- c. In the URL field, enter 192.168.1.254 and click Go. Envelopes (PDUs) will appear in the topology window.
- d. Minimize, but do not close, the HTTP Client configuration window.

• **Step 3:** Generate FTP traffic.

- a. Click FTP Client and open the Command Prompt from the desktop
- b. Enter the `ftp 192.168.1.254` command. PDUs will appear in the simulation window.
- c. Minimize, but do not close, the FTP Client configuration window.

• **Step 4:** Generate DNS traffic.

- a. Click DNS Client and open the Command Prompt.
- b. Enter the `nslookup multiserver.pt.ptu` command. A PDU will appear in the simulation window.
- c. Minimize, but do not close, the DNS Client configuration window.

• **Step 5:** Generate Email traffic.

- a. Click E-Mail Client and open the E Mail tool from the Desktop.
- b. Click Compose and enter the following information:
 - 1) To: `user@multiserver.pt.ptu`
 - 2) Subject: personalize the subject line
 - 3) E-Mail Body: personalize the Email
- c. Click Send.
- d. Minimize, but do not close, the E-Mail Client configuration window.

• **Step 6:** Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.

• **Step 7:** Examine multiplexing as the traffic crosses the network.

You will now use the Capture/Forward button in the Simulation Panel to observe the different protocols travelling on the network.

Note: The Capture/Forward button ‘ `**>|**` ’ is a small arrow pointing to the right with a vertical bar next to it.

- a. Click Capture/Forward once. All of the PDUs travel to the switch.
- b. Click Capture/Forward six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.

Questions:

What is this called?

R/ A esto se le llama multiplexación por tiempo a nivel de simulación visual.

A variety of PDUs appears in the event list in the Simulation Panel.

What is the meaning of the different colors?

R/ En la lista de eventos cada color representa un protocolo o tipo de mensaje, algunos de estos son:

Color	Significado
Celeste	HTTP request o tráfico de aplicación
Verde	TCP segment (capa de transporte)
Naranja	IP packet (capa de red)
Morado	Ethernet Frame (capa de enlace de datos)
Rojo	Error o mensaje ICMP ("Destination unreachable")
Amarillo	ARP request o reply

Part 2: Examine Functionality of the TCP and UDP Protocols

• **Step 1:** Examine HTTP traffic as the clients communicate with the server.

a. Click Reset Simulation.

b. Filter the traffic that is currently displayed to only HTTP and TCP PDUs. To filter the traffic that is currently displayed:

1) Click Edit Filters and toggle the Show All/None button.

2) Select HTTP and TCP. Click the red "x" in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only HTTP and TCP PDUs.

c. Open the browser on HTTP Client and enter 192.168.1.254 in the URL field. Click Go to connect to the server over HTTP. Minimize the HTTP Client window.

d. Click Capture/Forward until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.

Question:

Why did it take so long for the HTTP PDU to appear?

R/ Por que antes de enviar datos HTTP, TCP tiene que establecer la conexión mediante el three-way-handshake y eso incluye enviar y recibir diferentes segmentos con SYN.

e. Click the PDU envelope to show the PDU details. Click the Outbound PDU Details tab and scroll down to the second to the last section.

Questions:

What is the section labeled?

R/ La sección está etiquetada como TCP

Are these communications considered to be reliable?

R/ Si TCP es un protocolo orientado a conexión y confiable.

Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

R/ SRCPORT = 1027, DESTPORT = 80, SEQUENCE NUM = 1, ACK NUM = 1, estos valores van cambiando a cómo se transfieren los datos.

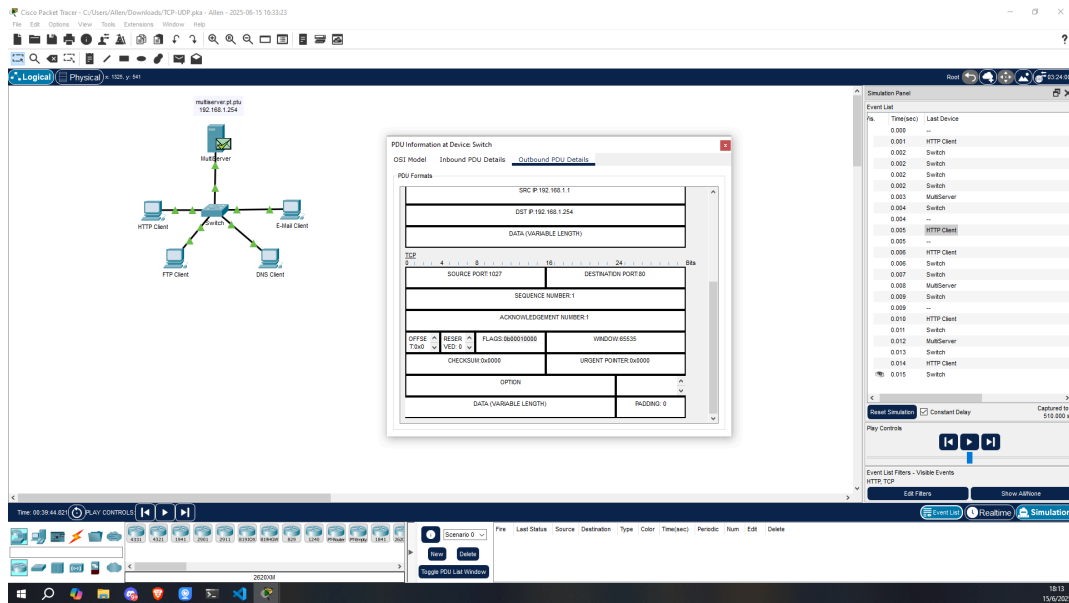
f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

Flag Place	6	5	4	3	2	1
Value	URG	ACK	PSH	RST	SYN	FIN

Question:

Which TCP flags are set in this PDU?

R/ En este PDU, la única flag activa es la 5, la de ACK



g. Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.

h. Click the PDU envelope and select Inbound PDU Details.

Question:

How are the port and sequence numbers different than before?

R/ Ahora el SRC PORT es 80 y el DEST PORT es 1027, el SEQUENCE NUM es 1 y el ACK NUMBER es 103

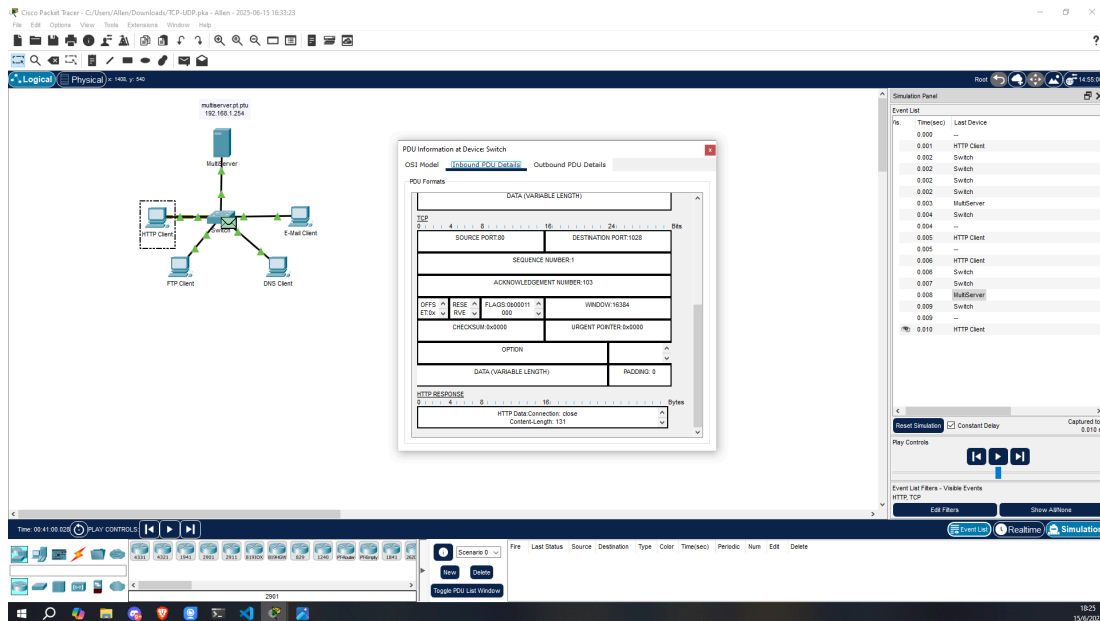
i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.

Question:

What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

R/ El SRC PORT es el dinámico del cliente y el DEST PORT es 80 del servidor, el SEQ NUM y el ACK NUM van a aumentar ya que la transferencia de datos continua correctamente, y además las flags de ACK y PSH estan activas ya que se esta enviando datos.

j. Reset the simulation.



• **Step 2:** Examine FTP traffic as the clients communicate with the server.

- Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering ftp 192.168.1.254.
- In the Simulation Panel, change Edit Filters to display only FTP and TCP.
- Click Capture/Forward. Click the second PDU envelope to open it.

Click the Outbound PDU Details tab and scroll down to the TCP section.

Question:

Are these communications considered to be reliable?

R/ Si, al igual que HTTP, FTP utiliza TCP que es fiable y orientado a conexión, lo que garantiza confiabilidad.

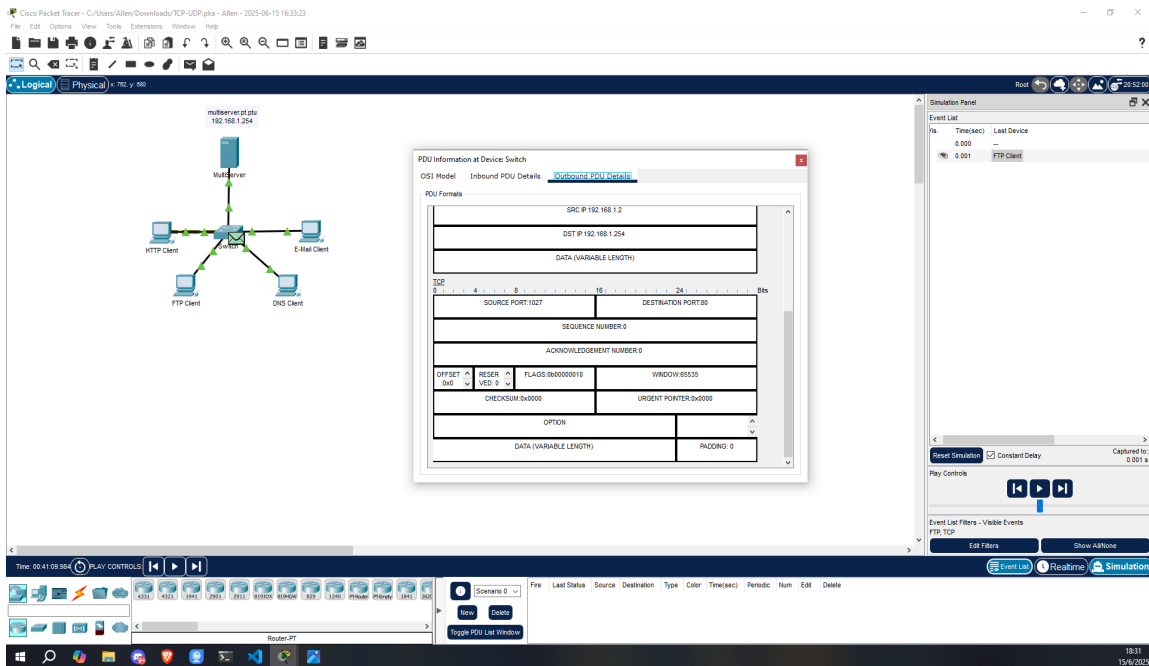
- Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

R/ El SRC PORT es 1027 y el DEST PORT es 80, el SEQUENCE NUM es 0 y el ACK NUMBER es 0

Question:

What is the value in the flag field?

R/ El único bit levantado en la flag es el que representa la bandera de SYN, representado el inicio del handshake.



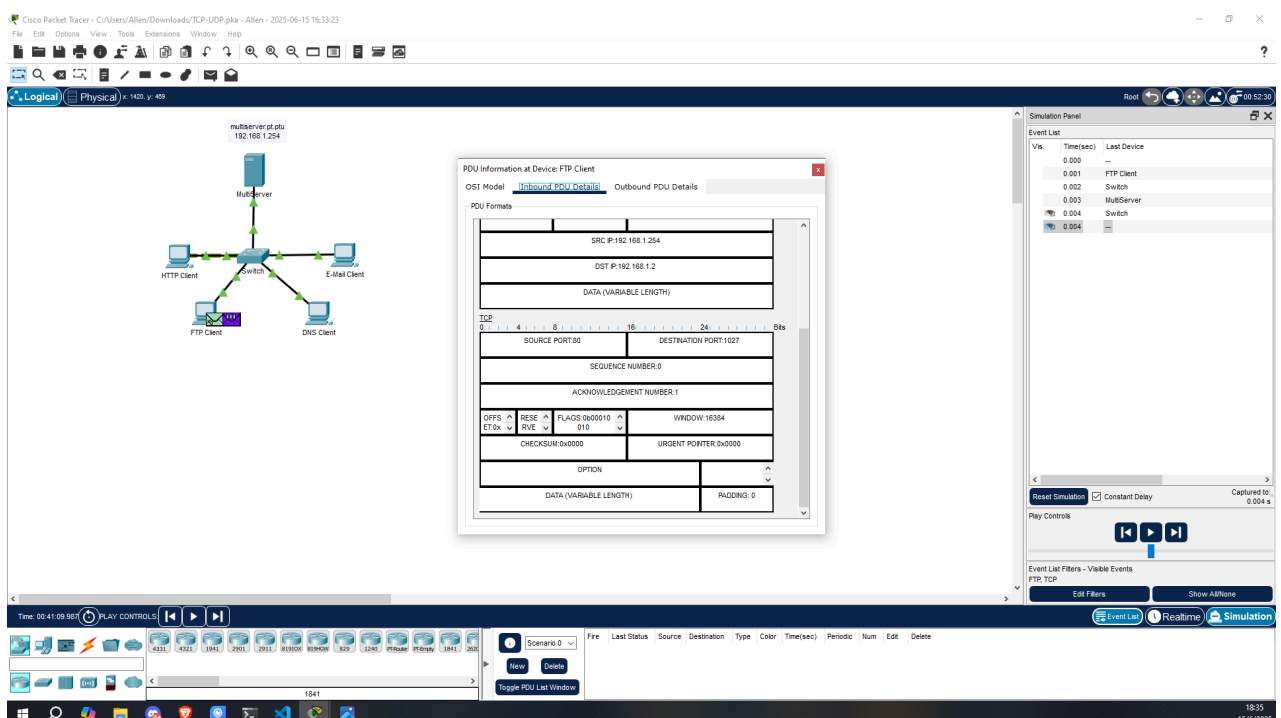
e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.

f. Click the PDU envelope and select Inbound PDU Details.

Question:

How are the port and sequence numbers different than before?

R/ Ahora el SRC PORT es 80 y el DEST PORT es 1027, el SEQUENCE NUM es 0 y el ACK NUMBER es 1 y ahora las banderas activa son las de SYN y ACK

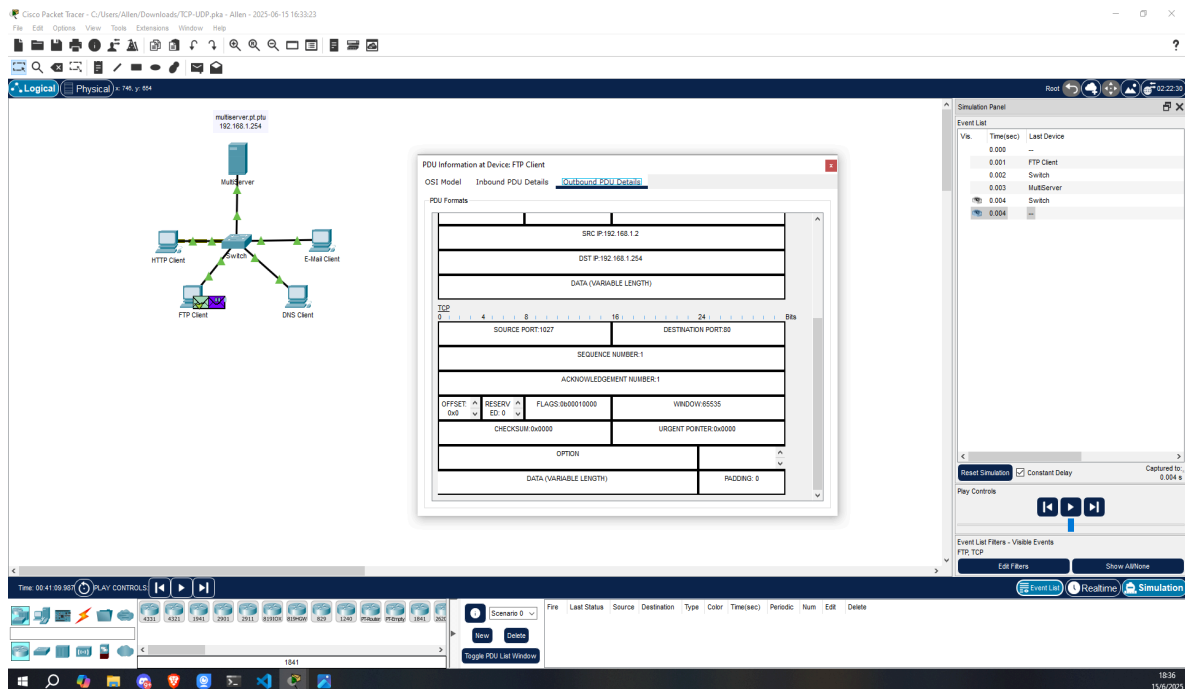


g. Click the Outbound PDU Details tab.

Question:

How are the port and sequence numbers different from the previous results?

R/ El SRC PORT es 1027, el DEST PORT es 80, el SEQUENCE NUM es 1 y el ACK NUM es 1, con la bandera de ACK activa



h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a different color.

i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.

Question:

What is the message from the server?

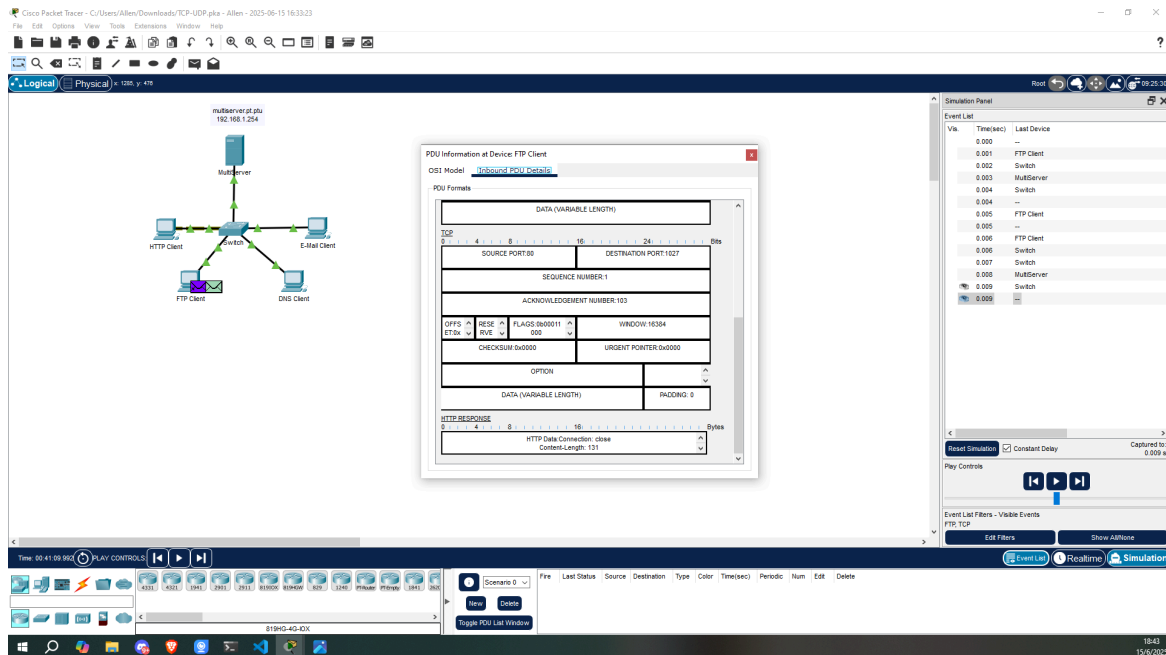
R/ El mensaje del servidor es :

"HTTP Data:Connection: close"

Content-Length: 131

Content-Type: text/html

Server: PT-Server/5.2"



j. Click Reset Simulation.

• **Step 3:** Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change Edit Filters to display only DNS and UDP.

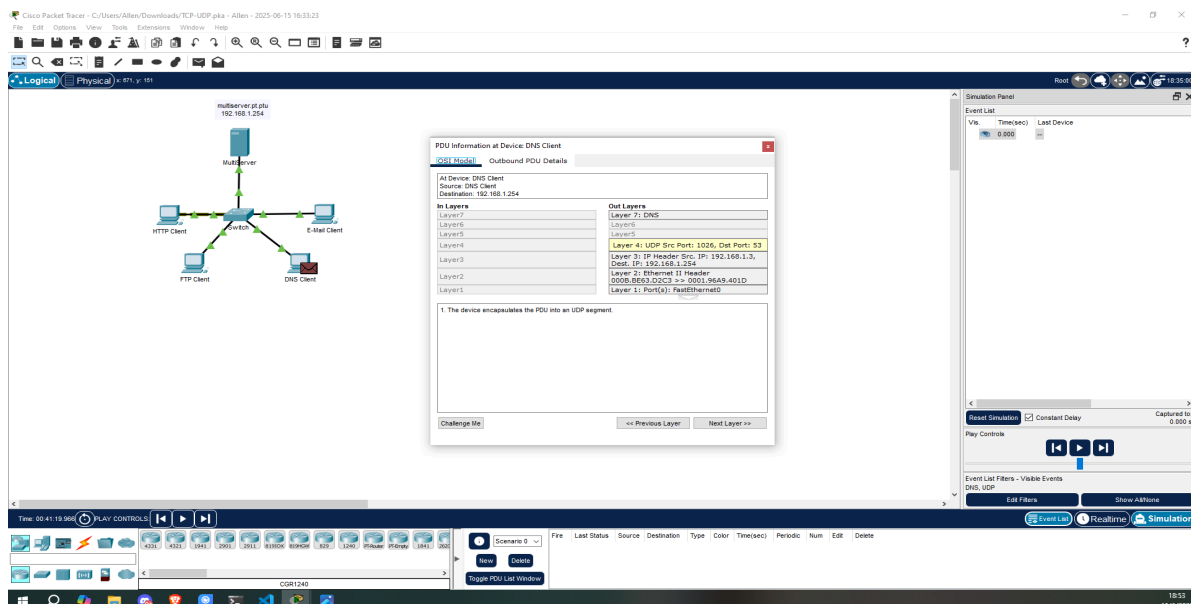
c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.

Question:

What is the Layer 4 protocol?

R/ En la capa 4 aparece UDP (User datagram protocol)

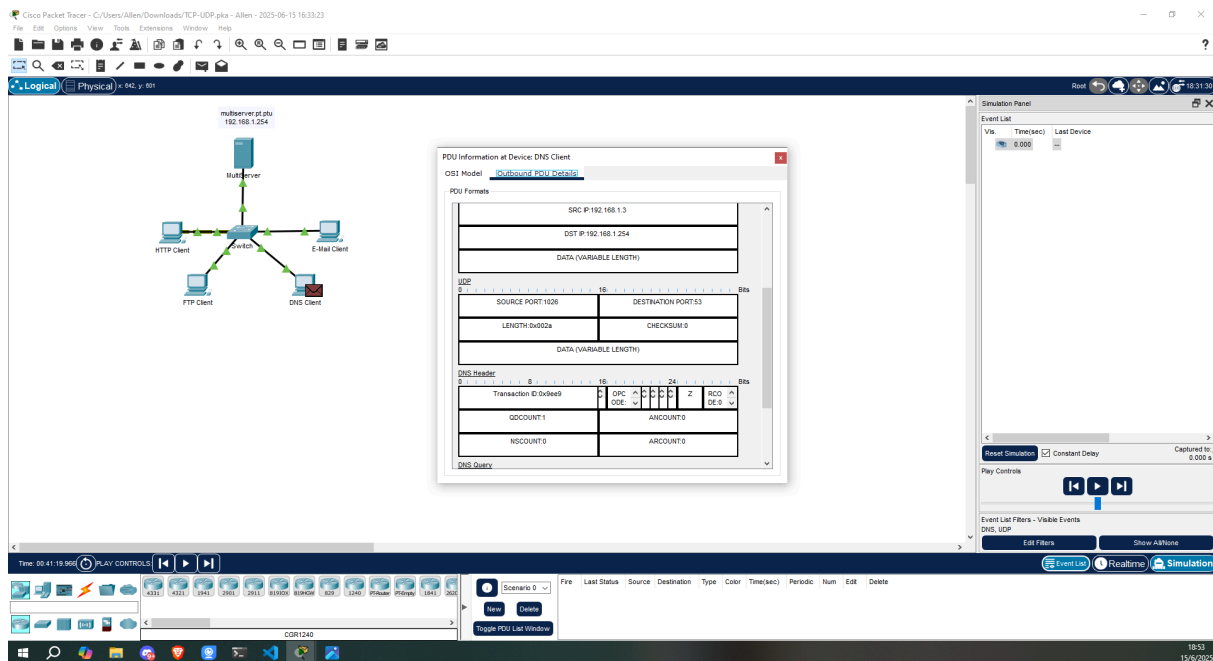


Are these communications considered to be reliable?

R/ No, UDP es un protocolo sin conexión, no es confiable, no tiene confirmaciones (ACK) y no tiene retransmisiones ni control de errores.

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC PORT and DEST PORT values.

R/ El SRC PORT es 1026 y el DEST PORT es 53



Question:

Why are there no sequence and acknowledgement numbers?

R/ Por que UDP no es orientado a conexión, no lleva control de orden o entrega de paquetes.

f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.

g. Click the PDU envelope and select Inbound PDU Details.

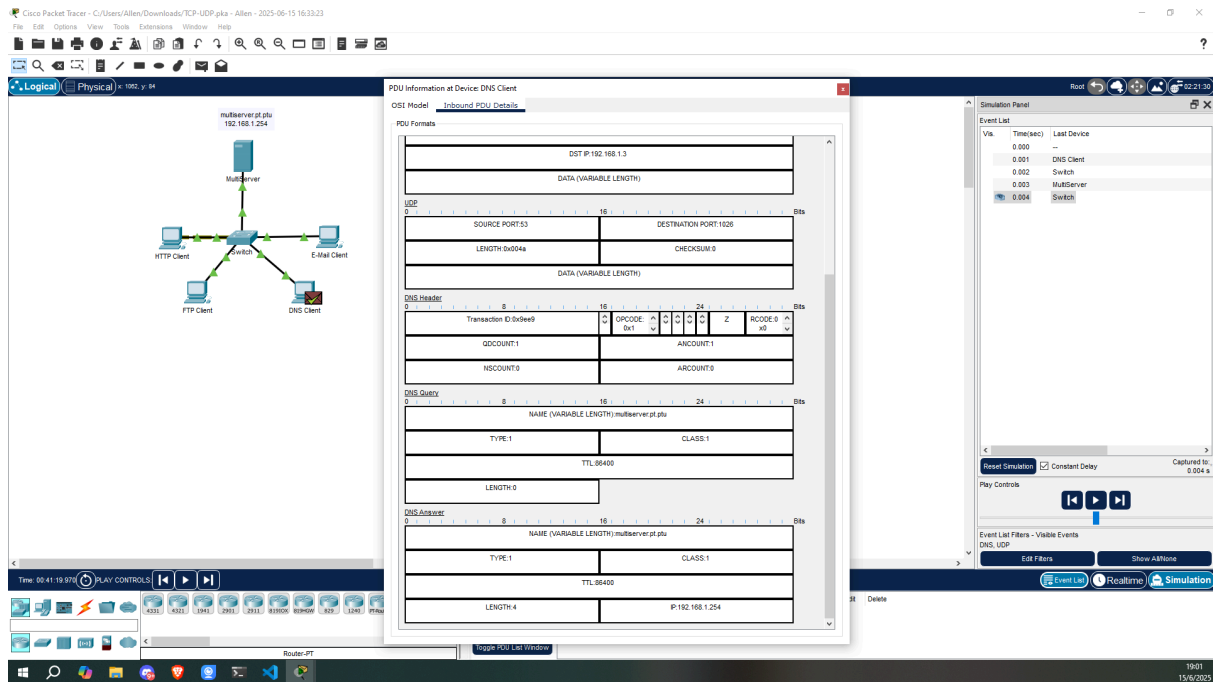
Question:

How are the port and sequence numbers different than before?

R/ En el PDU de vuelta, el SRC PORT es 53 y el DEST PORT es 1026

What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

R/ La IP address para el nombre multiserver.pt.ptu es IP:192.168.1.254



h. Click Reset Simulation.

• **Step 4:** Examine email traffic as the clients communicate with the server.

- Repeat the steps in Part 1 to send an email to user@multiserver.pt.ptu.
- In the Simulation Panel, change Edit Filters to display only POP3, SMTP and TCP.
- Click the first PDU envelope to open it.
- Click the Outbound PDU Details tab and scroll down to the last section.

Questions:

What transport layer protocol does email traffic use?

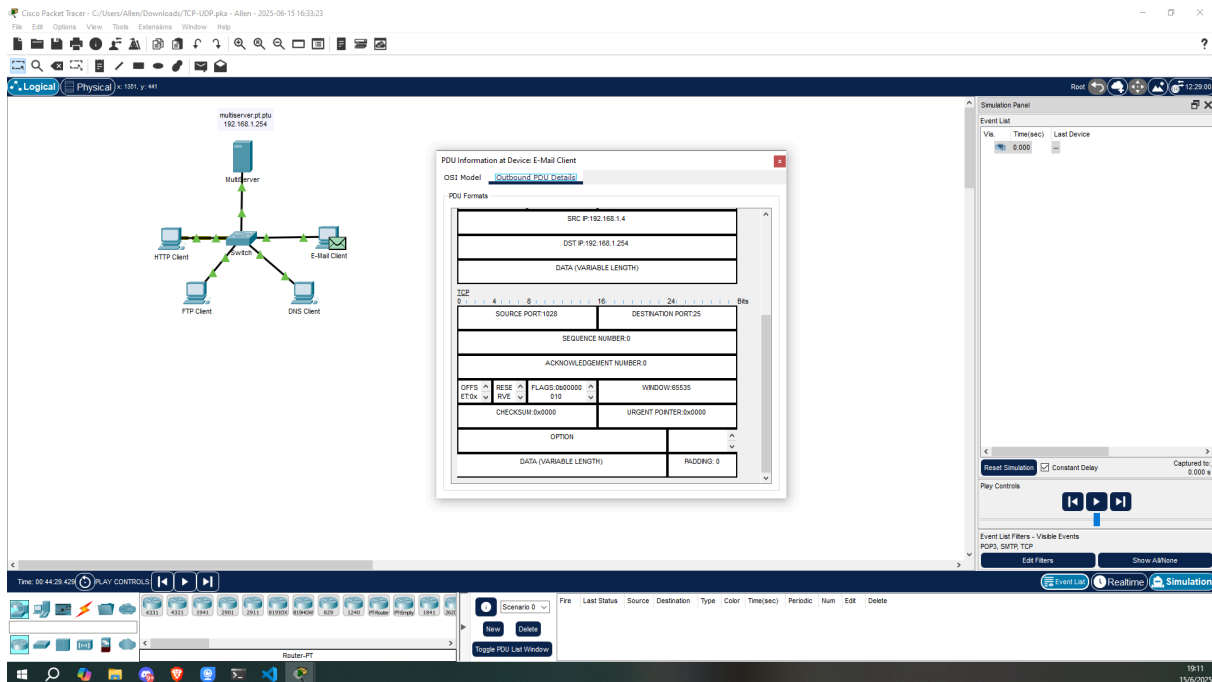
R/ El email traffic usa el protocolo TCP (Transport Control Protocol)

Are these communications considered to be reliable?

R/ Si TCP es un protocolo orientado a conexión y confiable.

- Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field value?

R/ El SRC PORT es 1028, el DEST PORT es 25, el SEQUENCE NUM es 0, el ACK NUM es 0 y en el flag field, el que aparece es el que indica la bandera de SYN, representado el inicio del handshake.



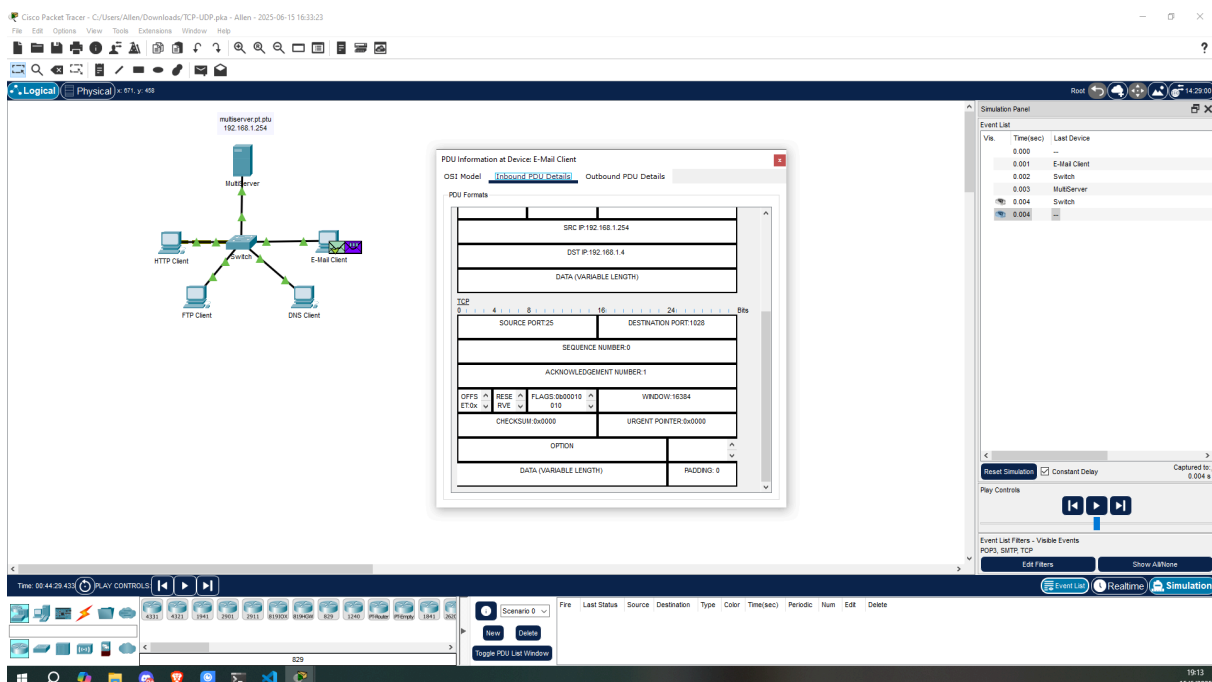
f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.

g. Click the TCP PDU envelope and select Inbound PDU Details.

Question:

How are the port and sequence numbers different than before?

R/ Ahora el SRC PORT es 25, el DEST PORT es 1028, el SEQUENCE NUM es 0, el ACK NUM es 1 y en el flag field, los que aparecen son los que indican la bandera de SYN y ACK.



h. Click the Outbound PDU Details tab.

Question:

How are the port and sequence numbers different from the previous two results?

R/ Ahora el SRC PORT es 1028, el DEST PORT es 25, el SEQUENCE NUM es 1, el ACK NUM es 1 y en el flag field, el que aparece activo es el bit de la flag ACK.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network diagram displays a central switch connected to a MultiServer (192.168.1.254) and four clients: HTTP Client, FTP Client, E-Mail Client, and DNS Client. The E-Mail Client is highlighted with a red envelope icon. In the center, the 'PDU Information at Device: E-Mail Client' window is open, showing the 'Outbound PDU Details' tab. The PDU details include: SRC P: 192.168.1.4, DST P: 192.168.1.254, DATA (VARIABLE LENGTH), TCP (SOURCE PORT: 1028, DESTINATION PORT: 25, SEQUENCE NUMBER: 1, ACKNOWLEDGEMENT NUMBER: 1, FLAGS: 080010, WINDOW: 65535, CHECKSUM: 0x0000, URGENT POINTER: 0x0000, OPTION, DATA (VARIABLE LENGTH), and PADDING: 0. On the right, the 'Simulation Panel' shows an 'Event List' with the following entries:

Time(sec)	Last Device
0.000	—
0.001	E-Mail Client
0.002	Switch
0.003	MultiServer
0.004	Switch
0.004	—

The 'Event List' also shows 'POP3, SMTP, TCP' as visible events. The 'Simulation' button is highlighted.

i. There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU Details.

Questions:

How are the port and sequence numbers different from the previous two PDUs?

R/ Ahora el SRC PORT es 1028, el DEST PORT es 25, el SEQUENCE NUM es 1, el ACK NUM es 1 y en el flag field, los que aparecen activos son los bits de la flag ACK y PSH.

The screenshot shows the Cisco Packet Tracer interface. On the left, the same network diagram as in the previous screenshot is displayed. In the center, the 'PDU Information at Device: E-Mail Client' window is open, showing the 'Outbound PDU Details' tab. The PDU details include: DATA (VARIABLE LENGTH), TCP (SOURCE PORT: 1028, DESTINATION PORT: 25, SEQUENCE NUMBER: 1, ACKNOWLEDGEMENT NUMBER: 1, FLAGS: 080011, WINDOW: 65535, CHECKSUM: 0x0000, URGENT POINTER: 0x0000, OPTION, DATA (VARIABLE LENGTH), and PADDING: 0. Below the TCP section, the 'SMTP DATA' section is visible, showing 'SMTP Data'. On the right, the 'Simulation Panel' shows an 'Event List' with the following entries:

Time(sec)	Last Device
0.000	—
0.001	E-Mail Client
0.002	Switch
0.003	MultiServer
0.004	Switch
0.004	—

The 'Event List' also shows 'POP3, SMTP, TCP' as visible events. The 'Simulation' button is highlighted.

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

R/ El puerto TCP 25 esta asociado al protocolo de email SMTP (Simple Mail Transfer Protocol) para enviar correos y el puerto TCP 110 esta asociado al protocolo POP3 (Post Office Protocol v3) para recibir correos.

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